Algorithms & Pseudocode

CSCI 1101 Introduction to Computer Science

Algorithm

Definition:

A set of finite rules or instructions to be followed in calculations or other problem-solving operations.

A Simple Algorithm

Examples:

Cookbook

Google Maps

Directions

Step 1

Whisk milk, eggs, vanilla, cinnamon, and salt together in a shallow bowl.

Step 2

Lightly butter a griddle and heat over medium-high heat.

Step 3

Dunk bread in the egg mixture, soaking both sides. Transfer to the hot skillet and cook until golden, 3 to 4 minutes per side. Serve hot.

Pseudocode

Definition:

A step-by-step description of an algorithm that does not use any programming language in its representation, but instead it uses the simple English language text.

Pseudocode is the **intermediate state between an idea and its implementation(code)** in a high-level language.

Pseudocode for Absolute Value Problem

Absolute Value:

```
Get Number
If Number < 0 then
    Multiply Number by -1
Output Number
```

Pseudocode for Is Positive Problem

Is Positive:

```
Get Number
If Number > 0 then
    Output "It's Positive"
Else
    Output "It's Not Positive"
```

Pseudocode Countdown Problem

Countdown:

```
Set Count to 3
While Count > 0 do
    Output Count
    Decrement Count by 1
```

Problem Solving Strategies

Ask Questions!

- What do I know about the problem?
- What information do I have to solve the problem?
- What does the solution look like?

Problem Solving Strategies

Reuse Solutions. Don't reinvent the wheel!

- Similar problems come up again and again.
- Recognize tasks that have been solved before and plug in the solution.

Problem Solving Strategies

Divide and Conquer!

- Break up a large problem down to smaller problems and solve for each smaller problem.
- You can continue to break down a problem over-and-over until each problem is manageable.